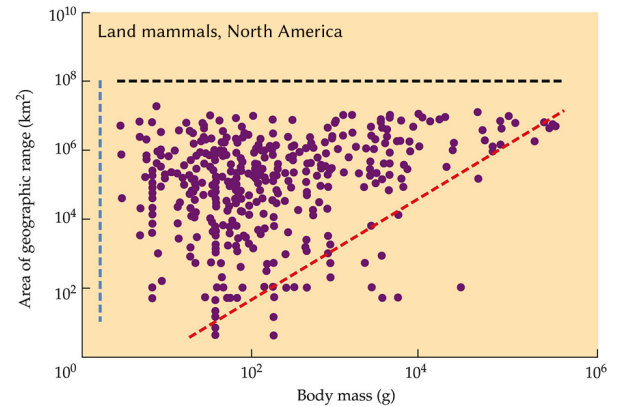


- (5 pts) 1. North American birds and mammals have various range sizes, from very small to very large. In contrast, Eurasian birds and mammals tend to have only very large range sizes. Explain the hypothesis that has been proposed to account for this discrepancy.

North America has mountain ranges that run north south that tend to act as barriers that limit the east-west range boundaries. Eurasia tends to have east-west running ranges so the east-west range boundaries are not restricted.

- (5 pts) 2. (a) Explain the relationship between body mass and geographic range size shown in the figure.

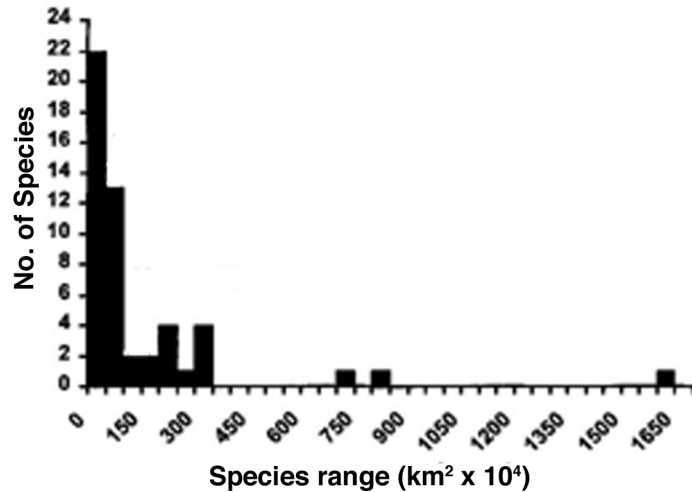


- (5 pts) (b) Explain why the range size for almost none of the mammals falls below the diagonal dashed line.

Answer here.

- (3 pts) 3. (a) This figure, based on African catarrhine primates, illustrates a common biogeographic pattern. In 1–2 sentences, describe what this figure illustrates.

Small geographic range size.



- (4 pts) (b) Explain the difference between the fundamental and realized geographic range.

The fundamental geo. range describes the geographic range that can be potentially occupied by a species. The realized range is the range actually occupied by a species after accounting for disturbance, competition, habitat quality, etc.

- (4 pts) (c) Explain how habitat quality can influence the range size of a species.

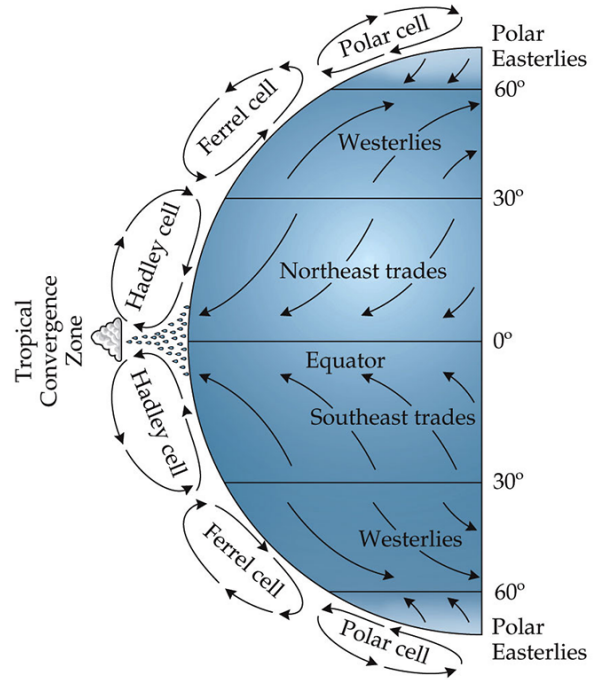
Habitat quality...

- (4 pts) (d) Explain how competition with other species can influence the geographic range size a species.

Competition...

- (12 pts) 4. Explain why arid regions like the Sahara Desert are concentrated around 30° latitude. Begin your explanation with solar input at the equator. You may use the figure as a guide.

Dry air descends, warms back up, sucks moisture out of the ground.



- (3 pts) 5. What physical geographic feature can also lead to desert formation? Name this effect.

Mountains create rainshadows.

- (3 pts) 6. (a) Draw and briefly state what is the latitudinal diversity gradient.

Species diversity is highest at low latitudes and decreases with increasing latitude.

- (5 pts) (b) Briefly explain how high net primary productivity contributes to the latitudinal gradient.

High NPP...

- (12 pts) (c) Use relative rates of origination, extinction, and immigration to explain the Out-of-the-Tropics concepts of cradles, museums, and immigration pumps. Tell how these concepts together could explain the latitudinal diversity gradient. You may include an illustration to support your explanation.

4 points each.

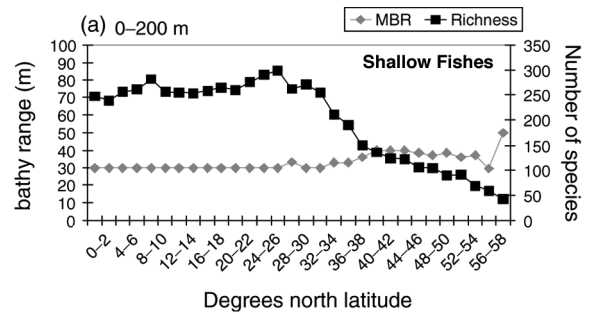
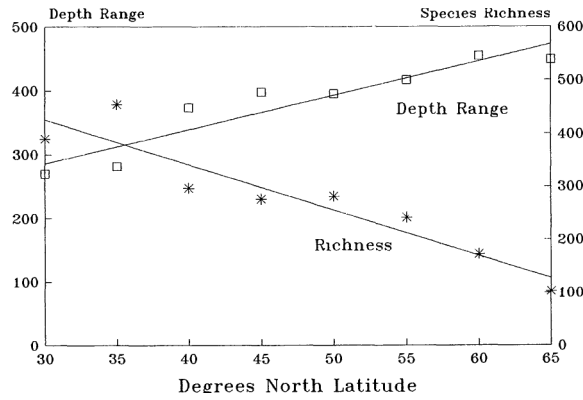
Cradles: $O_t > O_e$.

Museum: $E_t < E_e$.

Immigration pump: $I_t < I_e$.

Because origination is high and extinction is low in the tropics, species tend to accumulate in the tropics. Some taxa however do emigrate out of the tropics and immigrate into extra tropical regions. The tropics act as a source of taxa to extratropical regions but overall accumulation is highest in the tropics, leading to higher diversity in the tropics than in extratropical regions.

- (10 pts) 7. Stevens (1996) tested whether Pacific coastal fishes followed Rapoport's Rule. His results (below left) were consistent with the rule. Smith and Gaines (2003) analyzed Stevens' data in a different way. Smith and Gaines got different results than Stevens. One part of their results is shown below right. Comparing the two figures below, explain how the results of Smith and Gaines differed from Stevens for range size and richness at it relates to Rapoport's Rule.



Different bathymetric depths. Trend no longer applies.